Do Sanctions Break the Cycle of Violence?

Panel-VAR evidence on sanctions and organized violence, 1989–2019

Preliminary results — brief for discussion

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Summary for the meeting.

1. **Design.** A panel version of Jaeger and Paserman (2008): two reaction functions (new sanctions on past violence; violence on past new sanctions) forming a bivariate VAR(2) with state and year fixed effects, for 197 states, 1989–2019. Sanctions are *new* cases from the GSDB, violence is UCDP organized violence. Same outputs as the original: Granger tests, lag-structure grid, response functions, cumulative impulse responses.

2. **Senders react to violence.** Past violence predicts new sanctions (Granger $p = 0.007$ incidence, $p = 0.026$ levels, state and year FE only); a conflict year raises the number of new-sanction years over the next decade by 0.09–0.16 (11 of 12 IRF versions significant). The effect weakens when GDP growth is controlled for, which is plausibly a bad control.

3. **Violence reacts to sanctions with a two-year lag — in deaths, not in conflict incidence.** Two years after a new sanction case, log deaths per 100,000 are lower (sum of the lag coefficients -0.035 with fixed effects only, $p = 0.028$; -0.050 with controls, $p = 0.018$). This holds with every set of controls, with the jackknife, GMM, and PPML on winsorised deaths (-11 to -16%); the cumulative impulse response is negative but not significant. The 3.5 percentage-point lower conflict probability reported earlier appears only when the stock of sanctions in force in $t-1$ is controlled for. That stock contains the tested sanction lags, so this result is mechanical (Table 12). If sanctions may affect violence within the same year, the cumulative response turns positive.

4. **Sanctions coincide with escalations**: violence is already elevated before a new sanction and peaks in the sanction year. With annual data, who moves first within a year cannot be identified.

5. **Robustness.** Stationarity, stability and residual diagnostics are fine. GMM, PPML and raw (capped) deaths confirm the negative $t-2$ effect on deaths. All states, including the successor states of Yugoslavia, are in the main sample (only the Yugoslavia dummy was dropped from the controls); when these eight states are excluded as a robustness check, the effect is weaker ($p = 0.08$ with fixed effects only, Table 13). With GDELT (media-reported violence) the incidence estimate is negative but not significant, low-level

violence *rises* after sanctions, and senders do not react to GDELT violence.

6. **Decisions needed**: (i) main specification with or without controls (D8; proposal: fixed effects only, Table 12); (ii) how to handle within-year timing (monthly data?); (iii) whether the LP-DiD event study becomes part of the paper.

1 Introduction

Sanctions are the most frequently used coercive tool short of force, and a large share of them is imposed in response to political violence: to end civil wars, stop atrocities or punish repression. Whether they achieve this, or instead feed a cycle in which violence triggers sanctions and sanctions trigger more violence, is an open empirical question.

This project brings the empirical framework of Jaeger and Paserman (2008) — built to ask whether one side's violence predicts the other side's violence in the Second Intifada — to a cross-country panel: do senders react to violence in a target state, and does violence react to new sanctions? We estimate both reaction functions jointly as a panel VAR on 197 states over 1989–2019, using the Global Sanctions Data Base (Felbermayr et al., 2020; Yalcin et al., 2024) and UCDP data on organized violence (Davies et al., 2026).

Preliminary contribution.

- The first cross-country panel VAR of *sanctions* and *fatal violence*. The closest work, Dizaji and Murshed (2024), uses arms-import shocks as a proxy for embargoes and an expert risk index for conflict in 48 selected countries.
- A design that separates the two directions of the relationship, which the cross-sectional sanctions literature typically does not.
- Inference that takes the short-T panel seriously: Nickell-bias corrections, GMM, cluster bootstrap.

[Paper: motivation with recent examples (Myanmar 2021, Russia 2014/2022, Sudan); policy relevance; roadmap.]

2 Related literature

Sanctions and political violence. Most cross-national studies relate the *presence* of sanctions to an outcome of the target state. Economic sanctions are associated with more state repression (Wood, 2008), worse physical-integrity rights (Peksen, 2009) and declining democracy (Peksen and Drury, 2010). For armed conflict, the evidence is mixed. Hultman and Peksen (2017) study African civil conflicts 1989–2005 and find that threatened and imposed economic sanctions tend to intensify fighting, while imposed arms embargoes reduce it. Escribà-Folch (2010) finds that sanctions are associated with shorter civil wars, with institutional sanctions favouring negotiated settlements, whereas Lektzian and Regan (2016) argue that sanctions on their own rarely shorten civil conflicts and help mainly as part of a broader response including military intervention. At the sub-national level, Urtuzuastigui and Koren (2024) find that targeted sanctions on armed

actors increase violence against civilians (resource substitution); Stork (2020) reports a similar pattern for UN sanctions on rebel groups. The literature on targeted sanctions (Drezner, 2011; Biersteker et al., 2016) and on sanction success (Hufbauer et al., 2007) provides the institutional background. A common limitation is the treatment of sanctions as exogenous, although they are often imposed *because of* violence: this reverse direction is exactly what a VAR models.

Cycles of violence and VARs. Jaeger and Paserman (2008) estimate two reaction functions with daily fatality data and show that Palestinian fatalities Granger-cause Israeli responses but not the reverse; Asali et al. (2017) revisit the result with nonlinear VARs. The closest cross-country application is Dizaji and Murshed (2024), a panel VAR in which negative shocks to arms imports (a proxy for embargoes) are followed by lower military spending but *more* internal conflict; their methodological template is Dizaji and Farzanegan (2023). Their design differs from ours in four respects that matter: no sanctions data, an expert risk index instead of fatalities, 48 selected countries, and no treatment of the Nickell bias or of year effects.

Methods. Panel VARs go back to Holtz-Eakin et al. (1988); the GMM workflow we use as a robustness check follows Love and Zicchino (2006) and Abrigo and Love (2016). With $T \approx 30$, fixed-effects estimates of dynamic models are biased (Nickell, 1981); we use the half-panel jackknife (Dhaene and Jochmans, 2015), which Juodis et al. (2021) combine with a pooled Granger test for heterogeneous panels (implemented by Xiao et al., 2023); alternatives are Dumitrescu and Hurlin (2012) and, under cross-sectional dependence, Nazlioglu and Karul (2024). For impulse responses, VARs and local projections (Jordà, 2005) estimate the same object (Plagborg-Møller and Wolf, 2021), with a bias–variance trade-off (Li et al., 2024); panel local projections have their own Nickell bias (Mei et al., 2026). Event-study designs with on/off treatments (Dube et al., 2023; de Chaisemartin and D'Haultfœuille, 2026) are a possible extension. Finally, $\log(1+y)$ effects with many zeros are not percentage effects (Chen and Roth, 2024), which is why we read signs and significance from the levels specification and magnitudes from the incidence specification.

[**To do:** literature from the `papers/` folder is read in full; the political-science entries above are from abstracts — read in full before the paper version.]

3 Data

Sample. All states of the Gleditsch and Ward (1999) system with at least ten years of data in 1989–2019: 197 states and 5,932 state-years (four short-lived states excluded). States that are never in conflict (about 90) or never newly sanctioned (31) are kept: dropping them would select on the outcome. The panel is built with the `peacesciencer` framework (Miller, 2022).

Sanctions. The Global Sanctions Data Base, release 4 (Felbermayr et al., 2020; Yalcin et al., 2024), covers 1,547 cases 1950–2023. Because the stock of sanctions in force is highly persistent (more than half of all state-years are under some sanction), the VAR uses the *flow*: $\texttt{new_sanction}_{it} = 1$ if at least one case against state i begins in year t (incidence), and the number of new cases (levels). The construction reproduces the GSDB's own counts (e.g. 25 new and 38 active cases against Russia in 2022).

Violence. UCDP organized violence, country-year version 26.1 (Davies et al., 2026): deaths in state-based, non-state and one-sided violence. Incidence: conflict year with at least 25 deaths. Levels: log(1 + deaths per 100,000); the transformation is needed because the top 1% of state-years account for 99.9% of the variance of raw deaths (Rwanda 1994).

Controls (lagged one year): number of sanction cases in force, GDP per capita growth, V-Dem polyarchy (Coppedge et al., 2025).

Yugoslavia dummy (dropped). GSDB cases against Yugoslavia are attributed to its successor states (see `data/README_yugoslavia.md`), so one case can appear as a new case in several states in the same year. Earlier versions of this brief therefore also controlled for a dummy equal to one for the successor states of Yugoslavia in 1991–2016. With state fixed effects, however, this dummy is identified only from the years 2017–2019 in seven of these states (for Yugoslavia/FRY itself it is constant and fully absorbed), years with no new sanction and no conflict; and a level shift cannot remove cases that are counted in several states at once. It was dropped from the controls on 23 October 2026. Dropping it leaves the sanctions→violence tests unchanged (incidence $\chi^2(2) = 7.27$, $p = 0.026$; levels 8.08, $p = 0.018$) and moves the violence→sanctions tests only slightly (incidence 6.11 → 6.22, levels 2.79 → 2.82; Table 2, Panel A). The attribution problem itself is addressed in Section 7.5, which drops the ex-Yugoslav states or treats them as one cluster (Table 13).

Alternative violence measure. GDELT 1.0 (Leetaru and Schrodt, 2013): media-reported assault, fight and mass-violence events as a share of all GDELT events in the state-year (1979–2021), which captures violence below the UCDP threshold.

Table 1: Descriptive statistics, 1989–2019

Variable	Mean	SD	*N*
New sanction (0/1)	0.108	0.311	5,932
Number of new sanction cases	0.164	0.618	5,932
Conflict, ≥ 25 deaths (0/1)	0.205	0.403	5,932
log(1 + deaths per 100,000)	0.268	0.779	5,932
Deaths per 100,000 (raw)	4.541	164.641	5,932
Sanction cases in force, $t-1$	1.344	2.254	5,903
GDP per capita growth, $t-1$	0.018	0.041	5,874
V-Dem polyarchy, $t-1$	0.498	0.272	5,234

Notes: State-years of the estimation panel (197 states, 1989–2019). Sanctions: Global Sanctions Data Base (GSDB, release 4), flows of new cases constructed from case start years. Violence: UCDP organized violence, country-year (v26.1). Controls are lagged one year; missing controls are mostly microstates without V-Dem data.

[Paper: data appendix: case-level construction of flows, Yugoslavia attribution, GDELT country assignment; all decisions are documented in `data/README.md`.]

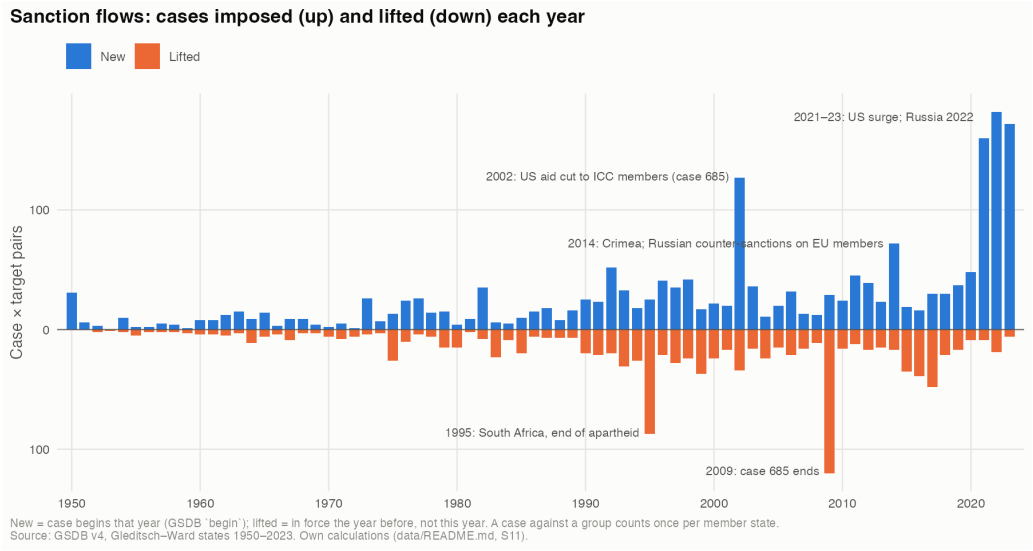


Figure 1: New and lifted sanction cases per year (GSDB)

4 Empirical strategy

Reaction functions. Following Jaeger and Paserman (2008), the two equations are

$$V_{it} = \sum_{k=1}^{p} a_k V_{i,t-k} + \sum_{k=1}^{p} b_k S_{i,t-k} + \gamma' X_{i,t-1} + \mu_i + \tau_t + \varepsilon_{it}, \tag{1}$$

$$S_{it} = \sum_{k=1}^{p} d_k S_{i,t-k} + \sum_{k=1}^{p} c_k V_{i,t-k} + \delta' X_{i,t-1} + \nu_i + \lambda_t + u_{it}, \tag{2}$$

with $p = 2$, state fixed effects (μ_i, ν_i), year fixed effects (τ_t, λ_t) and lagged controls X. Together they form a bivariate panel VAR. As in the original, each equation is estimated by OLS (linear probability models in the incidence version); standard errors are clustered by state.

Granger tests. H_0: $b_1 = b_2 = 0$ (sanctions do not predict violence) and H_0: $c_1 = c_2 = 0$ (violence does not predict sanctions); Wald $\chi^2(2)$. A rejection means predictive precedence conditional on the own history and the fixed effects, not causation.

Nickell bias. With $T \approx 30$ the fixed-effects estimates of dynamic coefficients are biased by $O(1/T)$ (Nickell, 1981), which is of the same order as their standard errors. We therefore report (i) the half-panel jackknife $\tilde{\beta} = 2\hat{\beta} - (\hat{\beta}_{1991-2005} + \hat{\beta}_{2006-2019})/2$ (Dhaene and Jochmans, 2015; Juodis et al., 2021), and (ii) a GMM panel VAR with forward orthogonal deviations and collapsed instruments (Abrigo and Love, 2016).

Impulse responses. The VAR traces the joint dynamics after a one-unit shock (one new sanction / one conflict year). Annual data leave the within-year ordering open, so we report three identifications: no same-year effect on the other variable (Jaeger and Paserman, 2008, fn. 12), and the two Cholesky orderings. Confidence bands come from a country-cluster bootstrap.

Response functions. The model-free analogue of Jaeger and Paserman (2008, Fig. 2): the difference between the mean outcome k years after an event year and in all other years, raw and net of state and year means.

[Paper: discussion of identification: what Granger causality can and cannot show here; forward-looking senders; omitted common drivers.]

5 Results

5.1 Granger tests

Table 2 is the panel analogue of Jaeger and Paserman (2008, Table 1); the full coefficients are in Table 15.

Sanctions → violence (Panel A). The lags of new sanctions are jointly significant at the 5% level in six of eight FE-OLS and jackknife versions (all eight at 10%) and in all four GMM versions. The effect sits at $t-2$. In levels, a new sanction case is followed two years later by fewer deaths per capita (sum -0.035 without and -0.050 with controls). For incidence, the sum is negative only with controls (-0.035, i.e. a 3.5 percentage-point lower conflict probability); without controls it is close to zero because the $t-1$ coefficient turns slightly positive. Table 12 (Section 7.5) shows which control is responsible: the stock of sanctions in force in $t-1$, which mechanically contains the new cases of $t-1$ and $t-2$. With the stock dated $t-3$, or with GDP growth or democracy alone, the incidence sum is again close to zero on the same sample. The incidence result should therefore not be read as an effect of sanctions; the levels result holds in every version.

Violence → sanctions (Panel B). Past violence predicts new sanctions in the specifications with fixed effects only (OLS and jackknife). With controls the test weakens, mainly through GDP growth: conflict lowers growth, and low growth predicts sanctions, so growth is plausibly a channel rather than a confounder. On a common sample, the incidence test is significant with every control and the levels test is lost only with GDP growth (Table 12). In GMM the incidence version is insignificant, but the GMM incidence models fail or nearly fail the Hansen test (Table 6).

5.2 Lag structure

Table 16 (appendix) repeats the tests for all combinations of one to three lags, as in Jaeger and Paserman (2008, Table 2). Senders react to the previous year's violence in every combination without controls. The sanctions→violence effect only appears once two lags of sanctions are included, consistent with an effect at $t-2$. BIC selects two lags for incidence and three for levels; two lags are the baseline.

5.3 Response functions

Figure 2 shows what happens around event years. Violence is already elevated two years before a new sanction, peaks in the sanction year and returns to its normal level after about two years. New sanctions rise in the year before a conflict year, peak in the conflict year and stay elevated

Table 2: Granger causality tests: panel version of Jaeger and Paserman (2008), Table 1

	FE-OLS		Half-panel jackknife		GMM	
	Controls	FE only	Controls	FE only	Controls	FE only
Panel A: Sanctions → violence						
Incidence	7.27^{**}	5.01^{*}	7.87^{**}	4.85^{*}	6.25^{**}	6.17^{**}
	(0.026)	(0.082)	(0.020)	(0.088)	(0.044)	(0.046)
	[−0.035]	[0.002]	[−0.040]	[0.006]	[−0.029]	[0.002]
Levels	8.08^{**}	7.14^{**}	7.84^{**}	6.18^{**}	11.52^{***}	7.48^{**}
	(0.018)	(0.028)	(0.020)	(0.045)	(0.003)	(0.024)
	[−0.050]	[−0.035]	[−0.054]	[−0.035]	[−0.068]	[−0.041]
Panel B: Violence → sanctions						
Incidence	6.22^{**}	9.86^{***}	3.93	11.29^{***}	0.52	0.56
	(0.045)	(0.007)	(0.140)	(0.004)	(0.772)	(0.758)
	[0.056]	[0.070]	[0.044]	[0.075]	[0.019]	[0.023]
Levels	2.82	7.30^{**}	3.86	22.78^{***}	2.95	7.77^{**}
	(0.244)	(0.026)	(0.145)	(<0.001)	(0.229)	(0.021)
	[0.031]	[0.045]	[0.037]	[0.092]	[0.034]	[0.074]
Observations	4,913	5,538	4,913	5,538	4,739	5,341
States	174	197	174	197	174	197

Notes: Wald $\chi^2(2)$ statistic for the joint significance of the two lags of the other variable in a bivariate panel VAR(2) with state and year fixed effects; p-value in parentheses; sum of the two lag coefficients in brackets (negative in Panel A = less violence after new sanctions). Incidence: new sanction (0/1) and conflict with $\geq$ 25 deaths (0/1), linear probability models. Levels: number of new cases and log(1+deaths per 100,000). Controls (lagged): sanction stock, GDP per capita growth, V-Dem polyarchy (the ex-Yugoslavia dummy of earlier versions is dropped, Section 3). FE-OLS and jackknife: standard errors clustered by state; jackknife $= 2\hat{\beta} - (\hat{\beta}_{1991-2005} + \hat{\beta}_{2006-2019})/2$ (Dhaene and Jochmans, 2015). GMM: forward orthogonal deviations, collapsed instruments (lags 2–5), year-demeaned; its Wald test ignores the covariance of the two lag coefficients (approximate). 1991–2019. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

for two more years. Sanctions arrive at the peak of escalations; whether the subsequent decline is an effect of the sanction or the end of the escalation cannot be read from this figure.

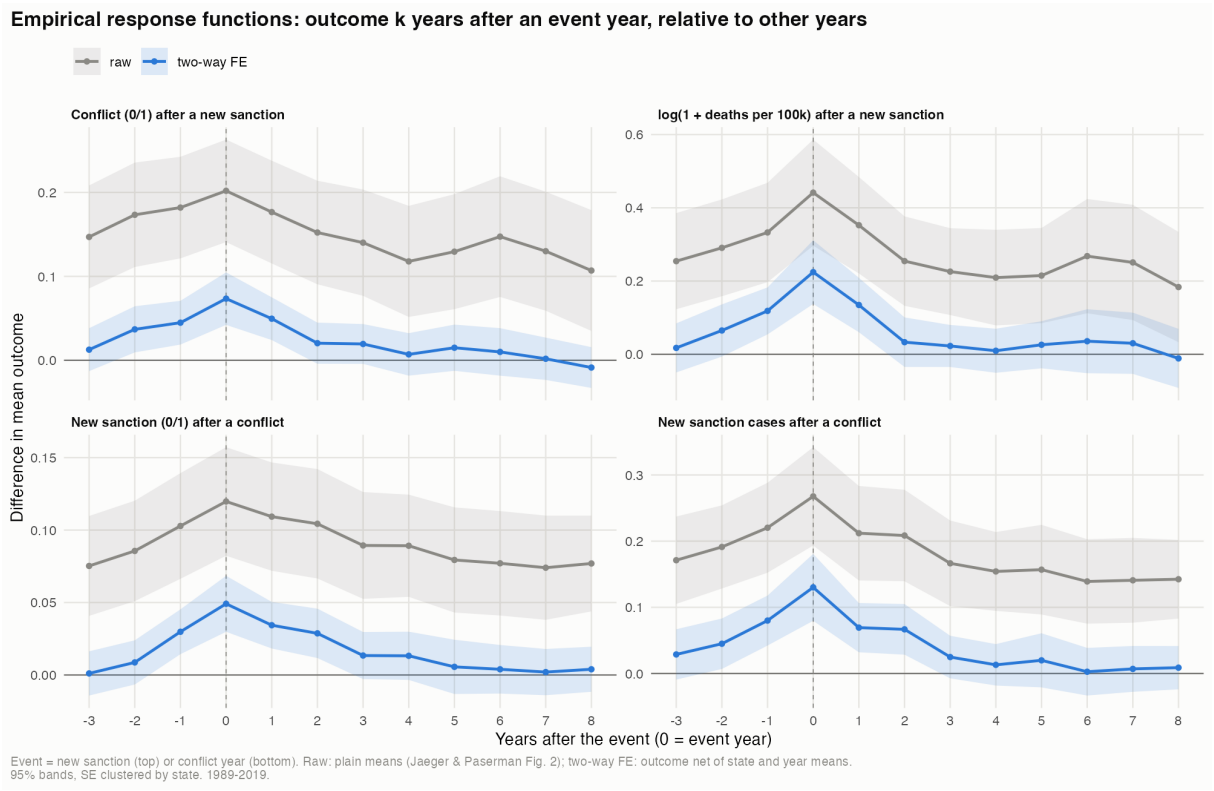


Figure 2: Empirical response functions (panel version of Jaeger and Paserman 2008, Fig. 2)

5.4 Impulse responses

Table 3 and Figures 3–4 report the dynamic responses.

- *Violence shock.* New sanctions respond in 11 of 12 versions. Allowing violence to trigger sanctions within the same year roughly doubles the response, so much of the sender reaction is fast.
- *Sanction shock.* Without a same-year effect, the cumulative response of violence is negative in three of four versions and significant for incidence with controls (-0.077 conflict-years over ten years) — a version that relies on the sanction-stock control (Section 7.5). If sanctions may affect violence within the year, the response turns positive and significant, because sanctions and violence spike in the same year.

All VARs are stable; the jackknife responses (dashed in Figure 3) decay more slowly, as expected once the downward Nickell bias in persistence is removed.

6 Diagnostics

Stationarity and persistence (Table 4). All four series reject a unit root in the panel tests, including CIPS, which allows for common shocks. Only states whose series varies can be tested; constant series (100 states never in conflict) are trivially stationary and stay in the sample.

Table 3: Cumulative impulse responses after ten years

		No same-year effect	Cholesky, sanctions first	Cholesky, violence first
Shock: new sanction; response: violence				
Incidence	with controls	-0.077^{*} [−0.154, −0.006]	0.031 [−0.072, 0.124]	-0.077^{*} [−0.154, −0.006]
Incidence	FE only	0.004 [−0.069, 0.083]	0.118^{*} [0.016, 0.223]	0.004 [−0.069, 0.083]
Levels	with controls	−0.150 [−0.300, 0.006]	0.177^{*} [0.023, 0.355]	−0.150 [−0.300, 0.006]
Levels	FE only	−0.107 [−0.256, 0.061]	0.239^{*} [0.061, 0.477]	−0.107 [−0.256, 0.061]
Shock: violence; response: new sanctions				
Incidence	with controls	0.121^{*} [0.039, 0.211]	0.121^{*} [0.039, 0.211]	0.215^{*} [0.107, 0.306]
Incidence	FE only	0.159^{*} [0.066, 0.260]	0.159^{*} [0.066, 0.260]	0.257^{*} [0.145, 0.369]
Levels	with controls	0.091 [−0.016, 0.218]	0.091 [−0.016, 0.218]	0.340^{*} [0.192, 0.496]
Levels	FE only	0.137^{*} [0.021, 0.260]	0.137^{*} [0.021, 0.260]	0.385^{*} [0.240, 0.541]

Notes: Sum of the responses over years 0–10 to a one-unit shock (one new sanction / one conflict year in the incidence VAR; one new case / one unit of log(1+deaths per 100,000) in the levels VAR). 95% country-cluster bootstrap intervals in brackets (199 draws); * interval excludes zero. No same-year effect: the identification of Jaeger and Paserman (2008, fn. 12); Cholesky orderings allow a same-year effect in one direction (rescaled to a unit own shock). For a sanction shock, "no same-year effect" coincides with "violence first"; for a violence shock, with "sanctions first".

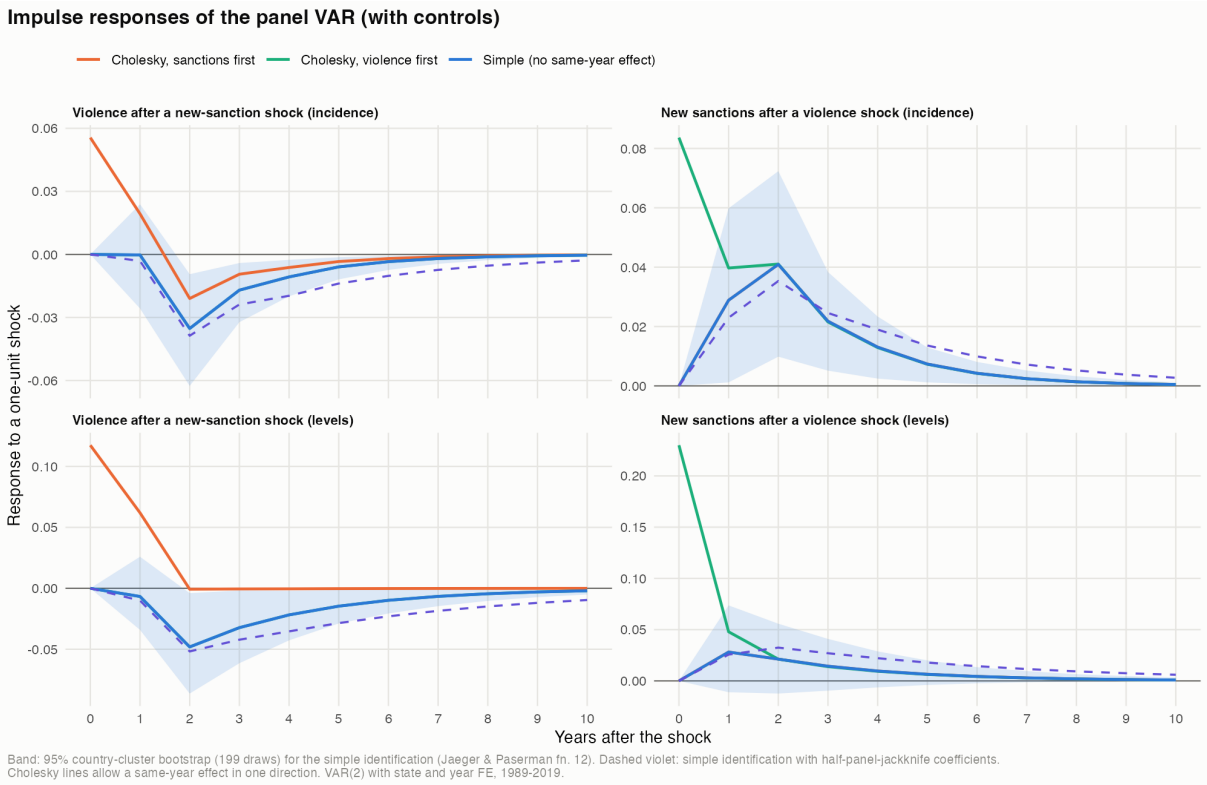


Figure 3: Impulse responses, VAR with controls

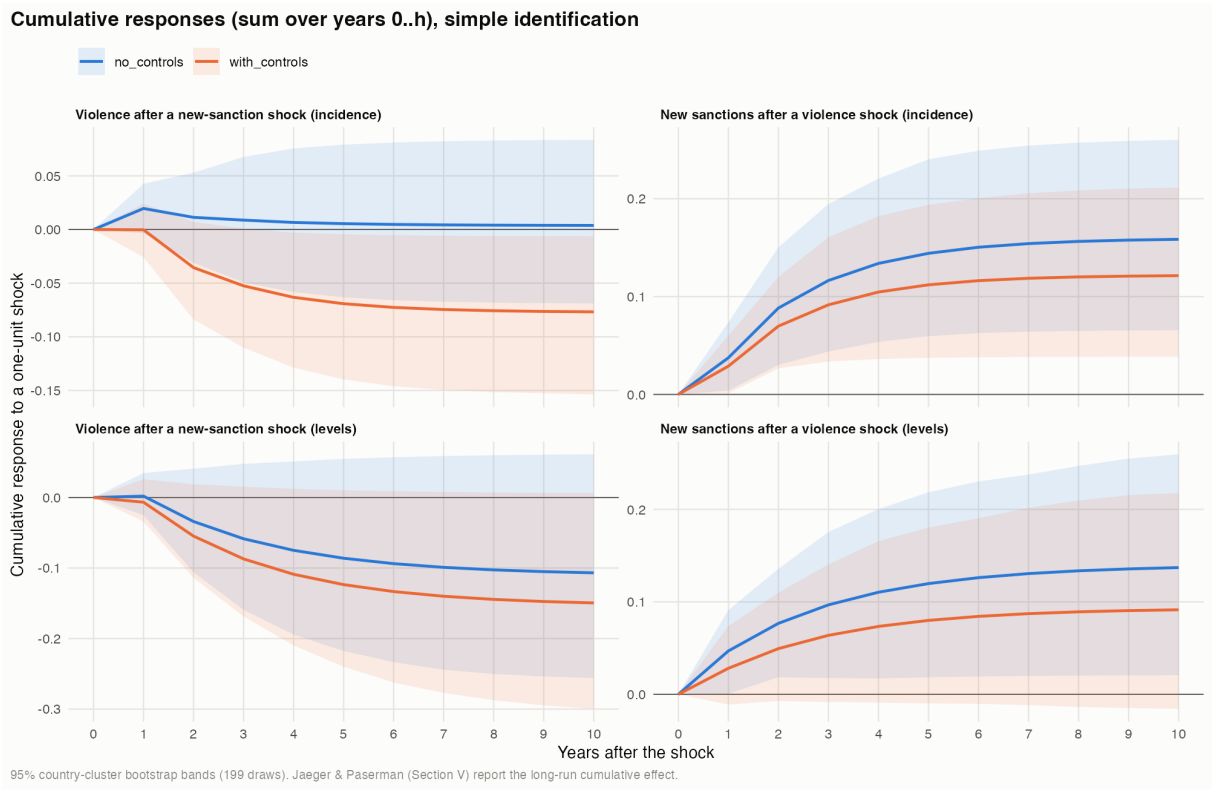


Figure 4: Cumulative impulse responses (no same-year effect)

Conflict is persistent (82% probability of continuing), new sanctions are close to events. The VAR is therefore estimated in levels.

Estimated models (Table 5). The violence equation has no remaining residual autocorrelation; the sender equation shows a very small one at lag 2 ($p \approx 0.04$), which a third own lag does not change in substance (Table 16). Residuals are weakly correlated across states (mean correlation 0.01–0.03), consistent with senders acting in waves; Driscoll–Kraay standard errors (Driscoll and Kraay, 1998) are planned as a check. All estimated VARs are stable (largest roots 0.56–0.80).

Table 4: Stationarity and persistence of the VAR variables

	Unit-root tests				Persistence		
Series	States tested	Share rejecting	Maddala–Wu, Choi p	CIPS (states)	FE AR(1), corrected	P(1|0)	P(1|1)
New sanction (0/1)	102	0.93	<0.001	−3.11 (84)	0.18	0.086	0.295
New sanction cases	102	0.92	<0.001	−3.30 (84)	0.12		
Conflict (0/1)	62	0.53	<0.001	−2.68 (52)	0.49	0.045	0.817
log deaths per 100k	97	0.61	<0.001	−3.02 (84)	0.70		

Notes: Unit-root tests, H_0: unit root in every state. State ADF regressions with intercept, 0–2 lagged differences by AIC; Maddala–Wu and Choi combine the state p-values (larger of the two shown); CIPS (Pesaran, 2007) adds cross-section averages against common shocks, balanced sub-panel, all $p < 0.01$. Only states whose series varies can be tested (constant series, e.g. never in conflict, are trivially stationary and stay in the sample). Persistence: FE AR(1) with state and year effects plus the Nickell correction $(1 + \rho)/(T - 1)$; P(1|0), P(1|1): probability of a 1 given last year's value. 1989–2019.

Table 5: Diagnostics of the estimated VARs

Model	Equation	Residual autocorr., p		Pesaran CD		Largest root		
		Lag 1	Lag 2	Mean corr.	p	OLS	JK	GMM
Incidence, with controls	Violence	0.503	0.688	−0.007	0.023	0.57	0.72	0.56
	Sender	0.077	0.042	0.010	<0.001			
Incidence, FE only	Violence	0.566	0.804	−0.005	0.089	0.60	0.75	0.57
	Sender	0.075	0.042	0.020	<0.001			
Levels, with controls	Violence	0.428	0.769	0.018	<0.001	0.67	0.80	0.67
	Sender	0.112	0.046	0.020	<0.001			
Levels, FE only	Violence	0.532	0.725	0.017	<0.001	0.69	0.80	0.69
	Sender	0.094	0.044	0.029	<0.001			

Notes: Residual autocorrelation: regression of the FE residual on its own lag, tested against the value $-1/(T-1)$ implied by within-demeaning (Wooldridge, 2010, section 10.5.4). Pesaran CD (Pesaran, 2015): states whose dependent variable varies (constant-outcome states have mechanically identical residuals); mean pairwise residual correlation and p-value. Largest root: modulus of the largest eigenvalue of the companion matrix (stable if < 1), FE-OLS, half-panel jackknife (JK) and GMM.

7 Robustness

7.1 GMM

GMM removes the Nickell bias by construction. It confirms the sanctions→violence results in all four versions (Table 2) and the levels results for violence→sanctions. The Hansen test does not reject in the levels models; it rejects for incidence without controls, so the GMM incidence results are the least reliable (Table 6).

Table 6: GMM panel VAR: specification tests

Specification	States	Obs.	Moments	Hansen J (df)	p	Largest root
Incidence, with controls	174	4,739	22	14.70 (8)	0.065	0.56
Incidence, FE only	197	5,341	16	16.50 (8)	0.036	0.57
Levels, with controls	174	4,739	22	8.31 (8)	0.404	0.67
Levels, FE only	197	5,341	16	7.60 (8)	0.473	0.69

Notes: Two-step GMM with forward orthogonal deviations and collapsed instruments (lags 2–5), variables demeaned by year (Abrigo and Love, 2016). Hansen J: H_0 instruments valid (Hansen, 1982); the number of moment conditions stays well below the number of states (Roodman, 2009).

7.2 Nonlinear models: logit and PPML

Why. The base-case reaction functions are linear: linear probability models for incidence and OLS on $\log(1 + \text{deaths per 100,000})$ for levels. Both choices are convenient but questionable with outcomes that are zero in most state-years. A linear probability model can predict outside $[0, 1]$ and imposes a constant marginal effect regardless of how likely conflict already is. A $\log(1 + y)$ outcome has no percentage interpretation when many observations are zero, and its coefficients change with the units in which y is measured (Chen and Roth, 2024). Jaeger and Paserman (2008, fn. 3) address the same concern in their data with probit and zero-inflated Poisson regressions.

Design. We re-estimate every reaction function of Table 2 with a model suited to its outcome, keeping the samples, the two lags, the controls, the state and year fixed effects and the clustering

by state unchanged (Table 7):

- *Logit* for the incidence equations (conflict 0/1, new sanction 0/1). To compare magnitudes with the linear model, the sum of the lag coefficients is converted into an approximate average partial effect on the probability.
- *Poisson pseudo-maximum likelihood* (PPML; Santos Silva and Tenreyro, 2006) for the levels equations, with the *untransformed* outcome: deaths per 100,000 and the number of new cases. PPML is consistent for any non-negative outcome with an exponential mean, handles zeros naturally and gives coefficients that are semi-elasticities. Because the death rate is extremely skewed (Rwanda 1994 alone has 12,577 deaths per 100,000), we also report a version with deaths winsorised at the 99th percentile (45.6).

With fixed effects, states whose outcome never changes carry no information and drop out: in the conflict logit 85–108 states with an unchanged conflict status (mostly states never in conflict), in the deaths PPML 61–84 states without any deaths, in the sender models 19–32 states never newly sanctioned.

Sanctions → violence. The pattern of the linear models holds in every version: the coefficient on the first lag of new sanctions is close to zero, the coefficient on the second lag is negative.

- *Logit, with controls:* the lags are jointly significant ($p = 0.040$); the average partial effect of -0.046 is close to the linear-probability estimate of -0.035, i.e. about a 4–5 percentage-point lower probability of conflict two years after a new sanction (mean 0.23). *Without controls,* the first lag turns positive and offsets the second ($p = 0.058$, average partial effect ≈ 0), exactly as in the linear model. As there, the incidence result with controls depends on the sanction-stock control (Section 7.5).
- *PPML on raw deaths:* the point estimates are large — death rates about a quarter to a third lower two years after a sanction — but imprecise ($p = 0.14$ with controls, 0.25 without). A Poisson likelihood weights observations by their size, so a handful of extreme state-years determine the estimate.
- *PPML on winsorised deaths:* once the extreme values are capped, the effect is precisely estimated ($p = 0.002$ with and without controls): death rates are 11–16% lower two years after a new sanction.

Violence → sanctions. Also unchanged. Without controls, past violence predicts new sanctions in the logit ($p = 0.006$; a conflict year raises the probability of a new sanction by about 6 percentage points over the next two years) and in the PPML ($p = 0.024$; 15% more new cases per unit of log deaths). With controls both weaken (logit $p = 0.065$, PPML $p = 0.46$), the same pattern as in the linear model, where GDP growth absorbs part of the sender reaction.

What the check adds and what it does not. The linear specification is not what drives the results: signs, lag pattern and significance survive a change of functional form, and the logit magnitudes are close to the linear ones. The PPML results also show that the levels finding does

not depend on the $\log(1+y)$ transformation; on the contrary, the semi-elasticities give the effect a natural scale (an 11–16% lower death rate). Three caveats remain.

1. *Different identifying sample.* Because states without variation drop out, the nonlinear models identify the effect from states that experience conflict at some point. The agreement with the linear models, which use all states, suggests that the never-violent states do not drive the linear estimates either; this is related to the conflict-prone subsample (decision D7).

2. *Incidental-parameter bias.* Fixed-effects logit with $T \approx 29$ has a bias of order $1/T$, the nonlinear counterpart of the Nickell bias; an analytical correction is a possible refinement. PPML with fixed effects does not have this problem for exogenous regressors, but the lagged outcome makes the regressors only predetermined.

3. *Sensitivity to extreme values.* The comparison of raw and winsorised PPML shows that the effect comes from the bulk of conflict state-years; the largest atrocities add noise rather than drive it. The cap at the 99th percentile is a choice, and other caps (95th, 99.5th) would be a natural check.

The functional-form check therefore strengthens the base-case reading: the small reduction in violence two years after a sanction, conditional on the controls, is not an artefact of linear models or of the log transformation; the same-year timing problem (Section 5) is unaffected by it.

Table 7: Robustness: logit and PPML versions of the reaction functions

		Linear (Table 2)		Nonlinear				
Model	Controls	$\chi^2(2)$	Sum	$\chi^2(2)$	Lag $t-1$	Lag $t-2$	Sum, comparable scale	States (dropped)
Sanctions → violence								
Logit	with controls	7.27**	−0.035	6.40**	0.023	−0.474	APE −0.046	89 (85)
Logit	FE only	5.01*	0.002	5.69*	0.322	−0.259	APE 0.007	89 (108)
PPML	with controls	8.08**	−0.050	3.95	−0.009	−0.402	−0.412 (−34%)	113 (61)
PPML	FE only	7.14**	−0.035	2.75	0.091	−0.383	−0.291 (−25%)	113 (84)
PPML, winsorised	with controls	8.08**	−0.050	12.56***	−0.005	−0.167	−0.172 (−16%)	113 (61)
PPML, winsorised	FE only	7.14**	−0.035	11.99***	0.023	−0.141	−0.117 (−11%)	113 (84)
Violence → sanctions								
Logit	with controls	6.22**	0.056	5.67*	0.297	0.226	APE 0.047	155 (19)
Logit	FE only	9.86***	0.070	10.21***	0.402	0.273	APE 0.058	165 (32)
PPML	with controls	2.82	0.031	1.60	0.011	0.061	0.072 (+8%)	155 (19)
PPML	FE only	7.30**	0.045	7.43**	0.127	0.011	0.137 (+15%)	165 (32)

Notes: Same samples, two lags, controls, state and year fixed effects and clustering as Table 2; fixed effects as dummies. Logit: incidence equations (conflict 0/1, new sanction 0/1); APE = sum of the two lag coefficients × mean $\hat{p}(1-\hat{p})$, an approximate average partial effect comparable to the linear-probability sum. PPML (Santos Silva and Tenreyro, 2006): levels equations with untransformed outcomes (deaths per 100,000, also winsorised at the 99th percentile; number of new cases), right-hand side as in the levels VAR; sum in log points and implied proportional change. States whose outcome never varies carry no information with fixed effects and are dropped. $^{*}p<0.1$, $^{**}p<0.05$, $^{***}p<0.01$.

7.3 Raw instead of log deaths

Jaeger and Paserman (2008) estimate their levels VAR on raw fatality counts. Our base case uses $\log(1+\text{deaths per 100,000})$ because the death rate is extremely skewed. Table 8 re-estimates the linear levels VAR with deaths per 100,000 instead, uncapped and winsorised at the 99th (45.6) and 95th (6.2) percentile of all state-years; the cap applies to the outcome and to its lags. Everything else is as in Table 2.

- *Uncapped raw deaths are not informative.* A single observation dominates the estimates: Rwanda's 12,577 deaths per 100,000 in 1994. It pulls the estimated persistence of violence down to 0.17 (0.67 with log deaths), and in the sender equation it produces an absurd Granger statistic ($\chi^2 \approx 300$), because Rwanda received two new sanction cases in 1996; without Rwanda the statistic falls to 4.7 ($p = 0.10$). This is the reason for the transformation in the base case.
- *Capped deaths: sanctions → violence holds.* With either cap, the lags of new sanctions are jointly significant with controls ($p = 0.036$ at p99, $p = 0.025$ at p95) and marginally without ($p \approx 0.055$), with negative sums: about 0.34 (p99) and 0.10 (p95) fewer deaths per 100,000 two years after a new case, i.e. roughly 17–24% of the mean of the capped variable.
- *Violence → sanctions* is significant without controls at the p99 cap ($p = 0.001$) and weaker with controls or with the stronger p95 cap, the same pattern as in the base case.

The base-case results therefore do not depend on the log transformation as long as the most extreme state-years are capped; the uncapped raw-count version, the analogue of Jaeger and Paserman (2008), is unsuitable for a cross-country panel that includes a genocide.

Table 8: Robustness: raw instead of log deaths in the levels VAR

	log(1+deaths) (Table 2)	Raw deaths	Capped at p99 (45.6)	Capped at p95 (6.2)
Sanctions → violence				
with controls	8.08**	0.55	6.63**	7.40**
	[−0.050]	[−2.535]	[−0.338]	[−0.096]
FE only	7.14**	4.18	5.68*	5.82*
	[−0.035]	[0.563]	[−0.186]	[−0.067]
Violence → sanctions				
with controls	2.82	292.54***	4.58	2.01
	[0.031]	[0.0001]	[0.0029]	[0.0169]
FE only	7.30**	323.10***	13.94***	4.32
	[0.045]	[0.0001]	[0.0048]	[0.0236]
Largest root (OLS)	0.67–0.69	0.17–0.18	0.57–0.58	0.69–0.70
Mean of violence variable	0.29	5.02	1.42	0.56

Notes: FE-OLS levels VAR(2) as in Table 2, with deaths per 100,000 instead of log(1+deaths per 100,000) as the violence variable (outcome and lags), as in the raw-count specification of Jaeger and Paserman (2008). Caps at the 99th and 95th percentile of all state-years 1989–2019 (deaths per 100,000 in parentheses). Granger $\chi^2(2)$; sum of the two cross-lag coefficients in brackets (sanctions→violence: deaths per 100,000 per new case; violence→sanctions: new cases per death per 100,000). Jackknife versions in `robustness_raw_deaths.csv`. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dynamics in deaths rather than log points. Because the capped version is still a linear VAR, it also delivers impulse responses in an interpretable unit, deaths per 100,000 — the analogue of the long-run numbers in Jaeger and Paserman (2008, Section V). Table 9 and Figure 5 use the p99 cap.

- *New sanction → deaths.* Without a same-year effect, one new sanction case is followed by 0.8 fewer deaths per 100,000 over ten years with controls (0.5 without), about half a year's average death rate (1.42). The sign matches the log version, but the bootstrap intervals

include zero: with deaths in levels, the dynamics are estimated less precisely than the Granger test. If sanctions may affect deaths within the same year, the cumulative response is positive (+1.1 to +1.6), as in the base case.

- *Deaths → new sanctions.* One additional death per 100,000 raises the number of new cases by about 0.01 over ten years (significant without controls), and by 0.03–0.04 if deaths may trigger sanctions within the same year (significant in all versions). Senders react, but per death the reaction is small.
- *Response function.* Capped deaths are already above normal in the year before a new sanction, peak in the sanction year (+1.8 deaths per 100,000 relative to the state's normal level) and are back to normal two years later — the same shape as with log deaths (Figure 2).

Table 9: Cumulative impulse responses after ten years, levels VAR with capped deaths

	No same-year effect	Cholesky, sanctions first	Cholesky, violence first
One new sanction case → deaths per 100,000			
with controls	−0.827	1.079*	−0.827
	[−1.985, 0.206]	[0.032, 2.330]	[−1.985, 0.206]
FE only	−0.465	1.600*	−0.465
	[−1.587, 0.609]	[0.358, 3.241]	[−1.587, 0.609]
+1 death per 100,000 → new sanction cases			
with controls	0.007	0.007	0.031*
	[−0.002, 0.016]	[−0.002, 0.016]	[0.017, 0.047]
FE only	0.012*	0.012*	0.036*
	[0.003, 0.022]	[0.003, 0.022]	[0.023, 0.053]

Notes: Levels VAR(2) with deaths per 100,000 capped at the 99th percentile (45.6) instead of log deaths; state and year fixed effects. Sum of the responses over years 0–10 to one new sanction case or to one additional death per 100,000; 95% country-cluster bootstrap intervals in brackets (199 draws); * interval excludes zero. Identifications as in Table 3.

Reading Figure 5. The three panels show the timing behind these numbers.

- *Left panel (deaths after a new case).* Nothing happens in the year after the sanction (cumulative response −0.01 with controls, +0.08 without). The decline starts in the second year (−0.33 and −0.21 deaths per 100,000 in that year alone), continues more weakly in the third, and has run its course after about five years, when the cumulative curve flattens at −0.8 (with controls) and −0.5 (without). This is the $t-2$ pattern of the Granger tests seen as a dynamic path. The bands are wide and include zero at every horizon, so the panel shows the direction and timing, not a precisely estimated effect.
- *Middle panel (new cases after one additional death per 100,000).* The sender reaction comes quickly: most of it in the first year after the violence shock (0.005–0.007 new cases), a little more in the second, and nothing after year three. Without controls the band excludes zero from the first year on; with controls it touches zero.
- *Right panel (deaths around a new-sanction year).* The grey line (raw means) is above zero at every horizon, including three years *before* the sanction: sanctioned states are more violent in

general, by 1–3 deaths per 100,000. Net of state and year means (green), deaths are at their normal level three and two years before a sanction, rise in the year before (+1.0), peak in the sanction year (+1.8), are still elevated one year later (+1.0) and are back to normal from the second year on, where they stay.

Read together, the left and right panels say the same thing as the log version: the decline that the VAR attributes to sanctions begins exactly when the escalation around the sanction year ends. Whether sanctions shorten that escalation or merely coincide with its end cannot be decided from annual data. What the figure does not show is any sign of a lasting *increase* in deaths after sanctions: two years on, deaths are back at the state's normal level, and in the VAR, which conditions on the sanction-year escalation, the point estimates are negative.

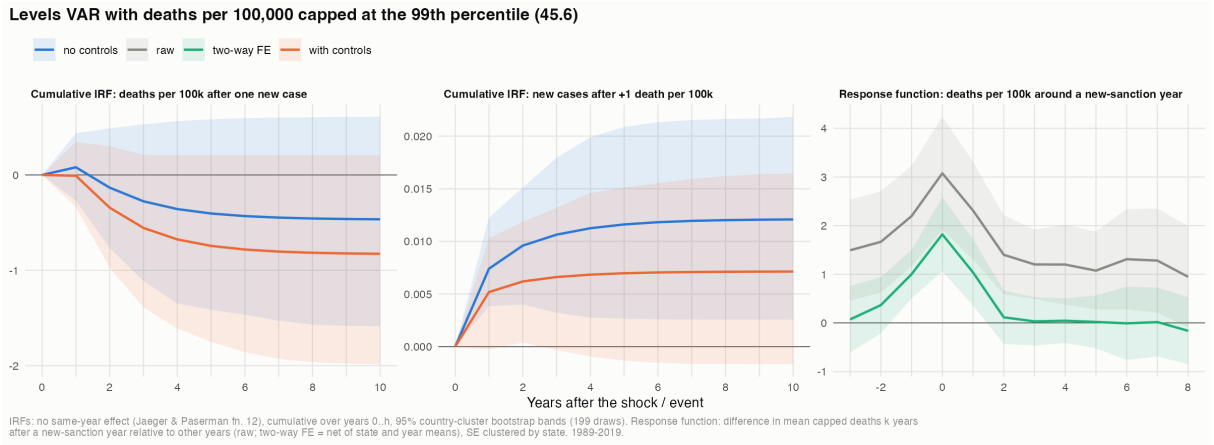


Figure 5: Levels VAR with deaths per 100,000 capped at the 99th percentile: cumulative impulse responses and response function

7.4 GDELT instead of UCDP

Table 10 repeats the Granger tests with media-reported violence.

- *Sanctions → violence.* Incidence with controls: a sum of lag coefficients of −0.032, almost identical to UCDP (−0.035) but not significant. Levels: positive and significant, i.e. more reported low-level violence after sanctions; the response functions show the violent share up by 0.5–0.7 percentage points in the sanction year and the year after (Figure 7).
- *Violence → sanctions.* Not found with GDELT. The GDELT incidence measure tracks UCDP conflict poorly, however (Table 11): a UCDP conflict year is above the state's GDELT median only 64% of the time. This is a weak test rather than a contradiction.

One possible reading: sanctions are followed by fewer deaths but more low-level unrest. This needs a better GDELT incidence definition before it can be claimed.

7.5 Controls and the Yugoslav cases

Which control drives what (decision D8). Panels A and B of Table 2 differ in two ways: the controls, and the sample (174 instead of 197 states, because V-Dem does not cover microstates). Table 12 separates the two by estimating every version on one common sample.

Table 10: Robustness: Granger tests with GDELT instead of UCDP violence

		UCDP	GDELT			
		1991–2019 OLS	1991–2019 OLS	1981–2021 OLS	≥100 events OLS	1991–2019 jackknife
Sanctions → violence						
Incidence	with controls	7.27** [−0.035]	3.84 [−0.032]	2.40 [−0.016]	4.21 [−0.032]	4.29 [−0.037]
Incidence	FE only	5.01* [0.002]	4.47 [0.010]	4.21 [0.021]	4.79* [0.010]	5.46* [0.023]
Levels	with controls	8.08** [−0.050]	7.19** [0.027]	4.18 [−0.025]	7.77** [0.027]	8.09** [0.039]
Levels	FE only	7.14** [−0.035]	11.18*** [0.143]	6.08** [0.097]	9.16** [0.091]	13.65*** [0.216]
Violence → sanctions						
Incidence	with controls	6.22** [0.056]	1.61 [−0.013]	0.36 [0.005]	1.36 [−0.012]	0.51 [−0.007]
Incidence	FE only	9.86*** [0.070]	0.45 [−0.006]	0.67 [0.007]	0.34 [−0.006]	0.10 [0.002]
Levels	with controls	2.82 [0.031]	1.26 [0.004]	1.81 [0.004]	1.15 [0.004]	6.14** [0.010]
Levels	FE only	7.30** [0.045]	2.14 [0.005]	2.59 [0.004]	2.18 [0.005]	12.78*** [0.013]

Notes: $\chi^2(2)$ Granger statistic, sum of the two lag coefficients in brackets. GDELT violence: assault, fight and mass-violence events per 100 GDELT events in the state-year (Leetaru and Schrodt, 2013); incidence = 1 if above the state's own 1979–2021 median. "≥100 events": state-years with fewer than 100 GDELT events set to missing. Units differ from UCDP, so compare signs and significance. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 11: How closely does GDELT track UCDP violence?

Statistic	Value
Correlation with UCDP log deaths, pooled	0.61
Correlation with UCDP log deaths, within states	0.47
Correlation with UCDP conflict (0/1), pooled	0.58
Correlation with UCDP conflict (0/1), within states	0.38
P(GDELT incidence = 1 | UCDP conflict year)	0.64
P(GDELT incidence = 1 | no UCDP conflict)	0.46
P(UCDP conflict | GDELT incidence = 1)	0.27

Notes: State-years 1989–2019 with both measures. GDELT measure: violent events in % of all GDELT events.

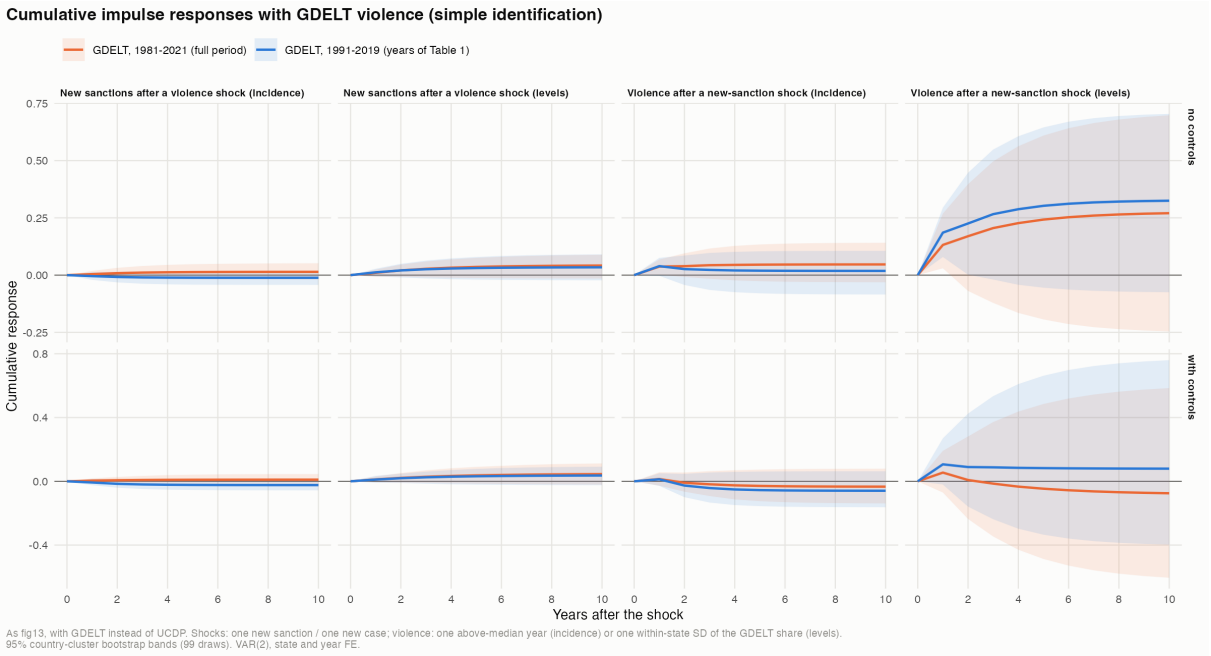


Figure 6: Cumulative impulse responses with GDELT violence

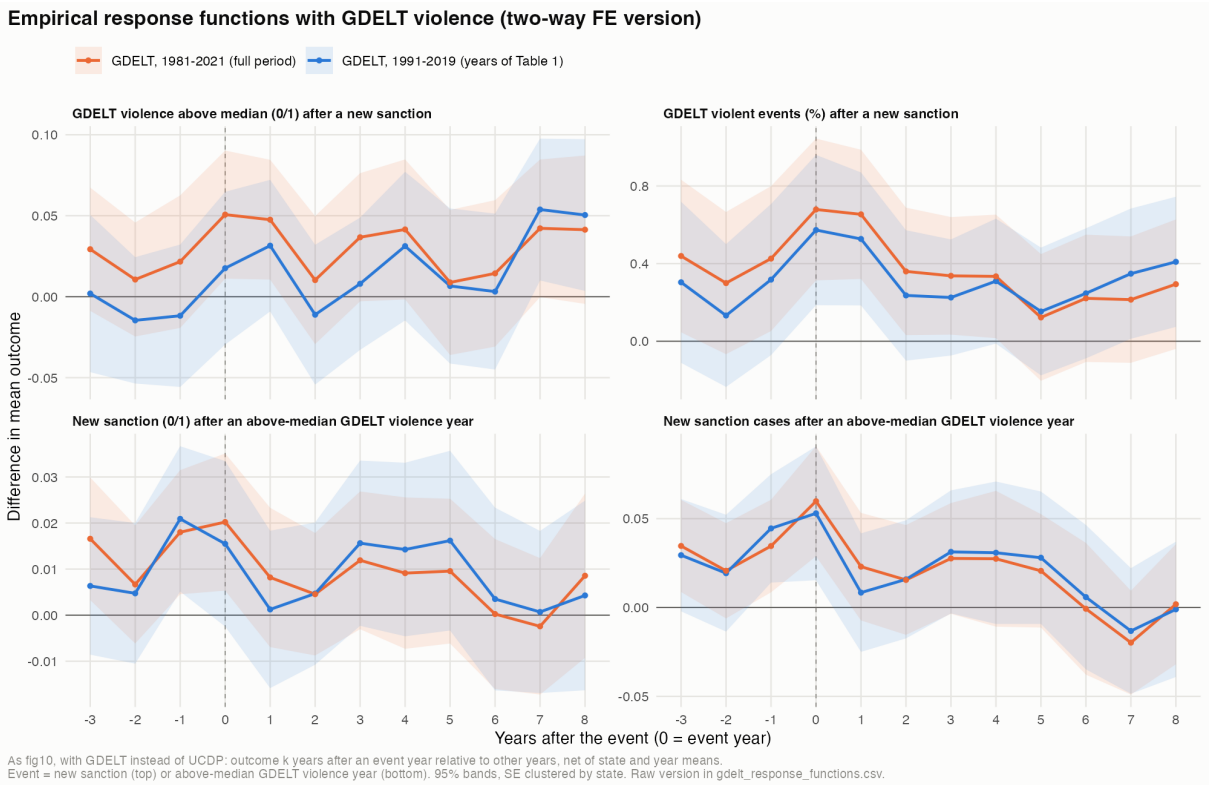


Figure 7: Response functions with GDELT violence

- *The sample does not matter:* fixed effects only on the 174-state sample give the same results as on the full sample.
- *Sanctions → violence, incidence:* the negative sum appears only with the stock of sanctions in force in $t-1$ (-0.032 alone, -0.036 with all controls); with fixed effects only, GDP growth, democracy, or the stock dated $t-3$, the sum is between -0.009 and 0.000. The stock in $t-1$ equals the stock in $t-3$ plus the new cases of $t-1$ and $t-2$ (minus lifted cases): it contains the regressors being tested. Conditioning on it changes what the sanction lags measure — a new case for a given number of cases still in force, identified mainly from cases that were lifted again quickly. This is a bad control, not evidence of an effect.
- *Sanctions → violence, levels:* -0.035 to -0.050, significant at 5% in every version.
- *Violence → sanctions:* incidence significant in every version; levels lose significance only with GDP growth.

Consequence: fixed effects only as the main specification; if a stock control is wanted, it should be dated $t-3$.

Table 12: Same-sample control check (decision D8)

	Incidence		Levels	
Controls	$\chi^2(2)$	Sum (SE)	$\chi^2(2)$	Sum (SE)
Sanctions → violence				
FE only	4.58	−0.000 (0.017)	7.05**	−0.035 (0.025)
Sanction stock, $t-1$	6.88**	−0.032 (0.020)	7.42**	−0.041 (0.025)
Sanction stock, $t-3$	5.29*	−0.000 (0.017)	7.02**	−0.036 (0.026)
GDP growth	4.15	−0.005 (0.018)	7.52**	−0.044 (0.026)
Democracy	4.46	−0.005 (0.018)	7.08**	−0.036 (0.026)
All controls (Table 2)	7.12**	−0.036 (0.020)	7.96**	−0.050 (0.026)
All controls, stock $t-3$	4.80*	−0.009 (0.018)	7.55**	−0.045 (0.027)
FE only, full sample (197 states)	5.01*	0.002 (0.017)	7.14**	−0.035 (0.025)
Violence → sanctions				
FE only	9.04**	0.067 (0.023)	6.76**	0.043 (0.020)
Sanction stock, $t-1$	9.16**	0.068 (0.023)	7.79**	0.049 (0.020)
Sanction stock, $t-3$	11.54***	0.076 (0.022)	7.97**	0.050 (0.020)
GDP growth	7.25**	0.060 (0.023)	2.97	0.031 (0.020)
Democracy	7.52**	0.061 (0.022)	5.73*	0.037 (0.020)
All controls (Table 2)	6.25**	0.056 (0.022)	2.79	0.030 (0.020)
All controls, stock $t-3$	7.81**	0.063 (0.022)	2.88	0.031 (0.020)
FE only, full sample (197 states)	9.86***	0.070 (0.022)	7.30**	0.045 (0.020)

Notes: FE-OLS VAR(2) with state and year fixed effects as in Table 2; all rows except the last on one common sample (4,888 observations, 174 states: all controls available, including the sanction stock in $t-3$). Sanction stock in $t-1$ = cases in force in $t-1$; it equals the stock in $t-3$ plus the new cases of $t-1$ and $t-2$ (minus lifted cases), i.e. it contains the tested sanction lags. Sum = sum of the two cross-lag coefficients. Standard errors clustered by state. $^{*}p<0.1$, $^{**}p<0.05$, $^{***}p<0.01$.

Yugoslav sanction cases. GSDB cases against Yugoslavia are attributed to its successor states, so one case can appear as a new case in several states in the same year. A period dummy for these states (used in earlier versions) is almost fully absorbed by the state fixed effects and does not address this; it was dropped from the controls. Table 13 instead drops the eight

ex-Yugoslav states, or treats them as a single cluster. Senders' reaction to violence is unaffected. The sanctions→violence effect is somewhat weaker without these states (levels, fixed effects only: -0.024, $p = 0.080$; with controls -0.040, $p = 0.044$), so part of it comes from the Yugoslav cases; clustering them together changes the p-values little.

Table 13: Robustness: attribution of Yugoslav sanction cases

		Incidence		Levels	
Version	Controls	$\chi^2(2)$	Sum (SE)	$\chi^2(2)$	Sum (SE)
Sanctions → violence					
Main (Table 2)	FE only	5.01^{*}	0.002 (0.017)	7.14^{**}	−0.035 (0.025)
Main (Table 2)	with controls	7.27^{**}	−0.035 (0.020)	8.08^{**}	−0.050 (0.025)
Without ex-Yugoslav states	FE only	3.96	0.010 (0.016)	5.06^{*}	−0.024 (0.022)
Without ex-Yugoslav states	with controls	5.30^{*}	−0.029 (0.019)	6.25^{**}	−0.040 (0.024)
Ex-Yugoslav states one cluster	FE only	4.91^{*}	0.002 (0.017)	5.35^{*}	−0.035 (0.025)
Ex-Yugoslav states one cluster	with controls	6.68^{**}	−0.035 (0.020)	6.35^{**}	−0.050 (0.025)
Violence → sanctions					
Main (Table 2)	FE only	9.86^{***}	0.070 (0.022)	7.30^{**}	0.045 (0.020)
Main (Table 2)	with controls	6.22^{**}	0.056 (0.022)	2.82	0.031 (0.019)
Without ex-Yugoslav states	FE only	9.13^{**}	0.069 (0.023)	6.30^{**}	0.046 (0.021)
Without ex-Yugoslav states	with controls	5.92^{*}	0.055 (0.023)	2.68	0.033 (0.020)
Ex-Yugoslav states one cluster	FE only	9.88^{***}	0.070 (0.022)	7.35^{**}	0.045 (0.020)
Ex-Yugoslav states one cluster	with controls	6.24^{**}	0.056 (0.022)	2.83	0.031 (0.019)

Notes: FE-OLS VAR(2) as in Table 2. GSDB cases against Yugoslavia are attributed to its successor states, so one case can appear in several states. Ex-Yugoslav states: Yugoslavia/FRY, Serbia, Montenegro, Macedonia, Croatia, Bosnia-Herzegovina, Kosovo, Slovenia. *One cluster*: standard errors clustered by state, except that the ex-Yugoslav states form a single cluster. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

7.6 Planned checks

- Leave-one-country-out estimates and the most influential sanction events.
- Bias-corrected heterogeneous Granger test (Juodis et al., 2021) on the balanced panel (167 states).
- Driscoll–Kraay standard errors.
- Time aggregation (two- and three-year periods), as in Jaeger and Paserman (2008, Table 3).
- Splits: sender (UN, EU, US), sanction type (arms embargoes), objective (conflict-related), period.
- Anticipation proxy (GDELT threats and coercion), analogous to Jaeger and Paserman (2008, Table 4).
- Sample to 2023; effects of sanction *lifting*.

8 Extension (not part of the base case): LP-DiD event study

The response functions of Section 5 compare event years with *all* other years. A modern alternative is the local-projection difference-in-differences estimator of Dube et al. (2023), which compares states with a new event to a set of "clean" control states and is designed for treatments

that switch on and off. This section describes what it estimates, what it shows, and how it relates to the VAR. It is kept separate from the base case because it answers a different question and its role in the paper is not yet decided.

8.1 Design

For each horizon $h = -3, \ldots, 8$ we estimate

$$y_{i,t+h} - y_{i,t-1} = \beta_h D_{it} + \gamma_h'(y_{i,t-1},\, y_{i,t-2}) + \tau_t + e_{it}, \tag{3}$$

on two groups of state-years only. *Treated*: an event in t and none in the three previous years. *Clean controls*: no event in t and none in the three previous years. In the sanction direction the event is a new sanction; in the reverse direction it is a conflict *onset* (a conflict year after three peaceful years), because conflict comes in spells. The outcome is the change relative to the year before the event, so time-invariant state differences drop out without state fixed effects; year effects absorb common shocks; the two lagged outcomes (used for $h \geq 0$) control for selection on recent violence, as in Dube et al.'s democracy application. The pre-event coefficients ($h = -3, -2$) are estimated without these controls, since otherwise $h = -2$ would be zero by construction; they test whether treated and control states were already moving differently before the year preceding the event.

Restricting comparisons to clean controls avoids the problem that two-way fixed-effects and panel local-projection regressions with on/off treatments implicitly use already-treated states as controls, which can bias them, even in sign (de Chaisemartin and D'Haultfœuille, 2026). Because the regression has no state fixed effects, the Nickell bias of panel local projections with fixed effects (Mei et al., 2026) does not arise in the same form.

8.2 Results

Table 14 and Figure 8 summarise the estimates.

- *Pre-trends.* In the UCDP data, treated and clean-control states move in parallel before the event in both directions (all pre-event coefficients small and insignificant).
- *Violence after a new sanction* (348 events). Relative to clean controls, the conflict probability rises by 8.6 percentage points in the sanction year and remains 6.3 points higher two years later; log deaths per capita rise by 0.23 and stay 0.10–0.12 higher up to eight years later (marginally significant at $h = 8$).
- *New sanctions after a conflict onset* (111 events). The probability of a new sanction rises by 12.8 percentage points in the onset year, with further increases at $h = 2$ and $h = 5$: senders react quickly and keep reacting while the conflict lasts.
- *GDELT.* The violent share of reported events is 0.8–1.1 percentage points higher in the years after a new sanction, with flat pre-trends. After a GDELT violence onset there is no sanction response, but the pre-trends are significantly negative, so this comparison is not credible with the median-based onset definition (Figure 9).

Table 14: LP-DiD event study (extension): change in the outcome relative to the year before the event

	UCDP: after a new sanction		UCDP: after a conflict onset		GDELT, 1991–2019	
Horizon h	Conflict (0/1)	log deaths per 100k	New sanction (0/1)	New cases	Violent share (%) after sanction	New sanction after onset
-3	−0.002	−0.020	0.004	0.032	0.307	−0.040
	(0.019)	(0.035)	(0.036)	(0.067)	(0.233)	(0.024)
-2	0.012	0.004	−0.005	0.002	0.043	−0.050**
	(0.017)	(0.034)	(0.038)	(0.063)	(0.169)	(0.023)
0	0.086***	0.230***	0.128***	0.427***	0.812***	0.010
	(0.019)	(0.054)	(0.036)	(0.142)	(0.179)	(0.019)
1	0.080***	0.199***	0.065*	0.148*	1.032***	−0.016
	(0.021)	(0.051)	(0.034)	(0.084)	(0.201)	(0.020)
2	0.063***	0.119**	0.124***	0.261***	0.710***	0.010
	(0.021)	(0.048)	(0.041)	(0.099)	(0.180)	(0.021)
3	0.040*	0.108**	0.038	0.040	0.836***	−0.008
	(0.021)	(0.046)	(0.035)	(0.054)	(0.225)	(0.022)
5	0.033	0.103**	0.107***	0.253**	0.586***	−0.007
	(0.023)	(0.044)	(0.041)	(0.115)	(0.214)	(0.018)
8	0.020	0.118*	0.025	0.028	0.930***	0.002
	(0.026)	(0.064)	(0.030)	(0.050)	(0.266)	(0.023)
Treated events	348	348	111	111	348	356

Notes: Coefficient on the treatment dummy in $y_{i,t+h} - y_{i,t-1} = \beta_h D_{it} + \gamma_h'(y_{i,t-1}, y_{i,t-2}) + \tau_t + e_{it}$ (Dube et al., 2023), estimated on treated state-years (event in t, none in the three previous years) and clean controls (no event in t and the three previous years); lagged outcomes only for $h \geq 0$; reference year $h = -1$. Conflict onset: conflict year after three years without conflict; GDELT violence onset: above-median year after three years at or below the median. GDELT columns 1991–2019. Standard errors clustered by state. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

8.3 How the event study relates to the VAR

At first sight the event study contradicts the VAR: violence is *higher* after a sanction here, while the VAR finds a *lower* conflict probability two years after a sanction. The two estimates condition on different things.

1. **Reference point.** The event study measures violence relative to the year *before* the sanction ($t-1$). In the violence equation of the VAR, the coefficient on a sanction in $t-2$ is estimated conditional on violence in $t-2$ and $t-1$ — that is, on violence in the *sanction year* itself and the year after. The VAR therefore asks whether violence after the sanction year is lower than the sanction-year level and its usual persistence predict; the event study asks whether it is higher than before the sanction. Both can be true at the same time: sanctions come with an escalation, after which violence declines faster than usual (VAR) but stays above its pre-sanction level (event study). The response functions (Figure 2) show exactly this shape.

2. **Within-year timing.** The jump at $h = 0$ mixes the escalation that led to the sanction with any effect of the sanction itself. Flat pre-trends only show that treated states were not on a different path up to $t-1$; they cannot rule out that the sanction responded to violence earlier in the same year. This is the same identification problem as the ordering of the impulse responses, and the event study does not solve it.

3. **Total versus partial effect.** The event study measures everything that follows a new sanction episode, including follow-up sanctions (P(new sanction | new sanction last year) = 0.30) and the conflict dynamics already under way. The VAR isolates the response to a

sanction innovation holding the past of both variables fixed.

4. **Comparison group.** Clean controls are states without any new sanction for four years; senders did not target them, and they may differ systematically from targeted states (for example, be more peaceful). The year effects and lagged outcomes reduce but do not remove this selection; in the reverse direction, onsets of conflict are compared with persistently peaceful states.

The direction of reverse causality (violence → sanctions) is the robust common finding: sanctions follow conflict onsets in the event study, Granger-follow violence in the VAR, and rise around conflict years in the response functions.

8.4 Assessment and possible next steps

Strengths: transparent, model-free comparison with a clear treated/control structure; robust to the negative-weighting problem of fixed-effects regressions; explicit pre-trend test; easy to communicate. *Weaknesses for our question:* it cannot separate within-year reverse causality, it measures a total effect that includes follow-up sanctions, and its controls are selected on not being sanctioned. Possible refinements, if it enters the paper:

- Condition on the escalation: compare sanctioned states with clean controls that had the same violence in the event year (i.e. add y_{it} or match on conflict status in t), which brings the estimand close to the VAR's.
- Restrict to conflict-prone states (in conflict in $t-1$), the subsample where sanctions are typically imposed (decision D7 in the analysis plan).
- Separate sanctions by objective (conflict-related) and by type (arms embargoes), where an effect on violence is most plausible.
- Report the equally-weighted (reweighted) version of Dube et al. (2023) and vary the clean-control window.

[**To do:** decide: (a) appendix as a complement to the response functions, (b) main-text robustness section, or (c) drop.]

9 Discussion and next steps

1. **Main specification (D8).** Fixed effects only. The only result that needs the controls — the incidence effect of sanctions on violence — comes from the sanction stock in $t-1$, which contains the tested lags (Table 12); the violence→sanctions result is weakened by GDP growth, plausibly a bad control. Controls (with the stock dated $t-3$) as robustness.
2. **Within-year timing.** The sign of the sanction effect depends on whether sanctions can affect violence in the year they are imposed. Monthly data (UCDP GED events and sanction announcement dates) would resolve this and bring the design closer to the daily data of Jaeger and Paserman (2008).
3. **Measurement.** A stricter GDELT incidence definition (top quartile, mass-violence events).

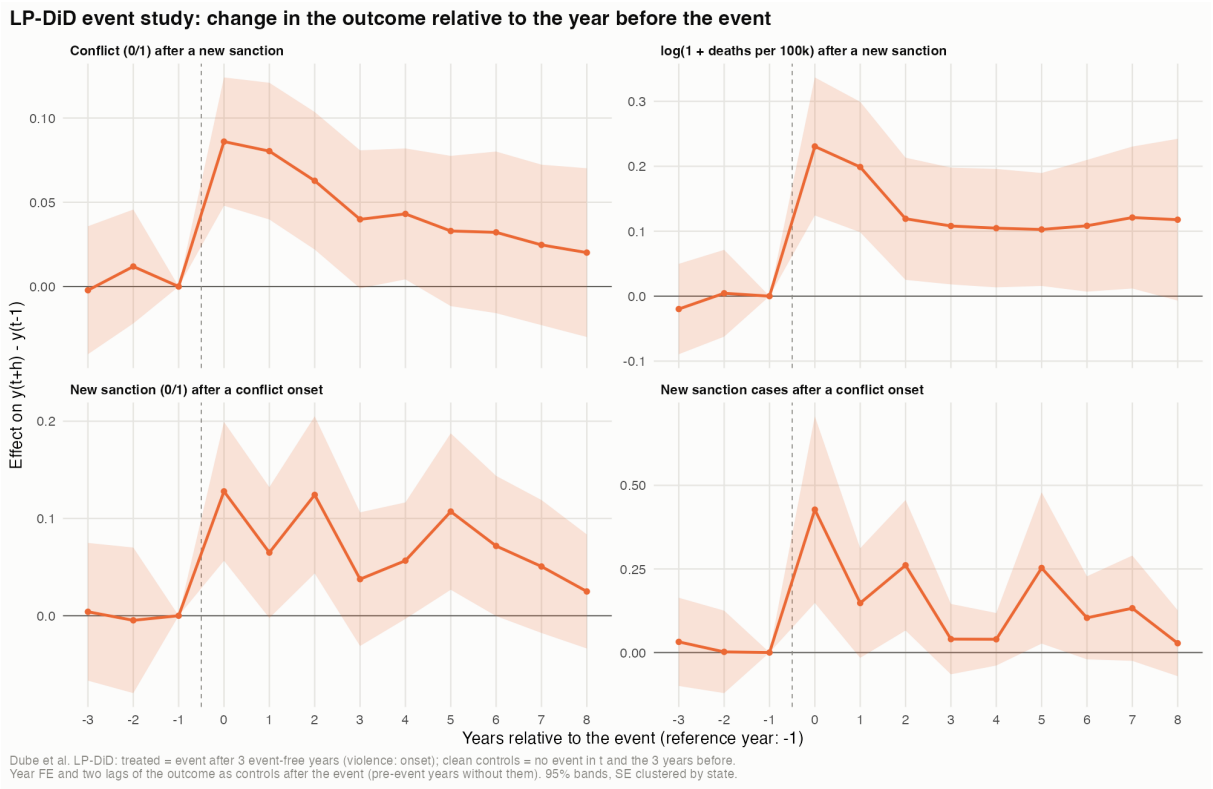


Figure 8: LP-DiD event study with UCDP violence (extension)

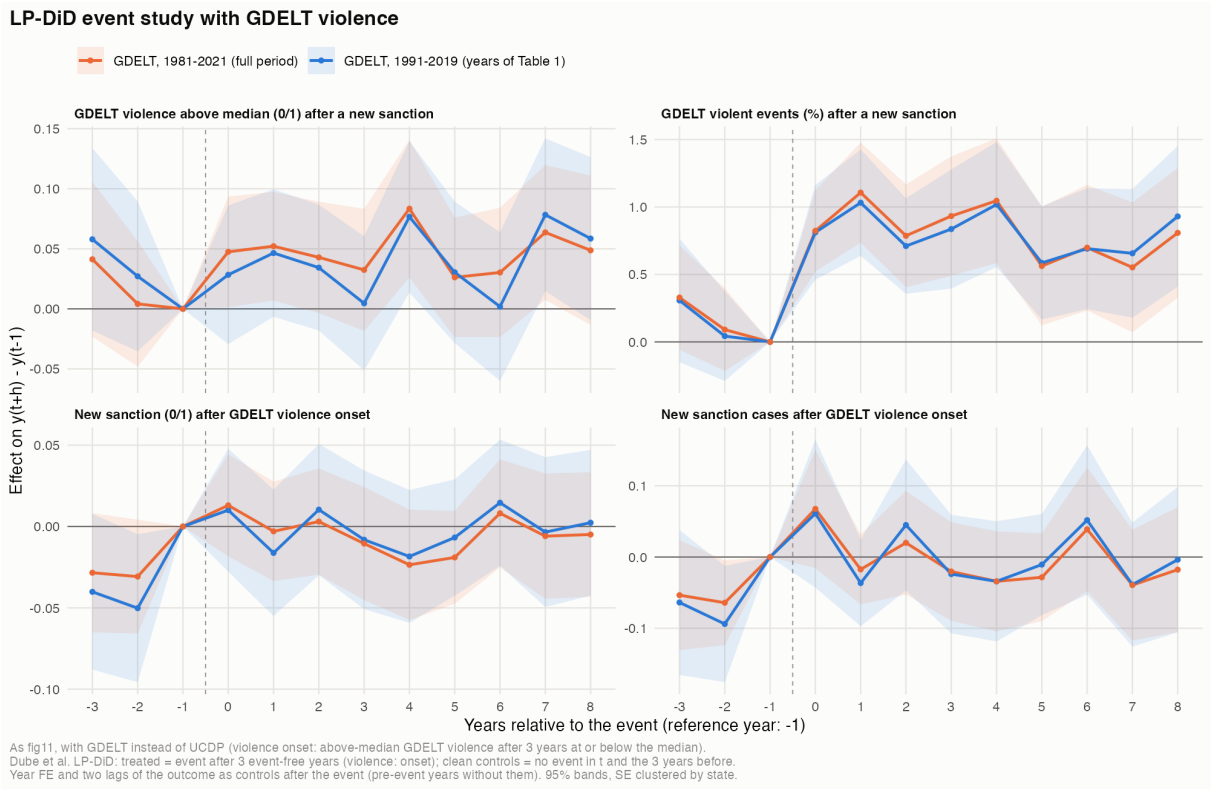


Figure 9: LP-DiD event study with GDELT violence (extension)

4. **Functional form.** Logit, PPML and raw-deaths (capped) versions confirm both directions (Section 7); no further work needed here beyond a bias-corrected logit.

5. **Interpretation.** Rather than a symmetric cycle of violence, the evidence so far suggests that senders respond to violence, while sanctions are followed by somewhat fewer deaths two years later — partly driven by the Yugoslav cases and with a sign that depends on timing assumptions — but not by a lower probability of conflict, and possibly by more low-level violence.

   [Paper: conclusion.]

References

Michael R. M. Abrigo and Inessa Love. Estimation of panel vector autoregression in Stata. *Stata Journal*, 16(3):778–804, 2016.

Muhammad Asali, Aamer Abu-Qarn, and Michael Beenstock. The cycle of violence in the Second Intifada: Causality in nonlinear vector autoregressive models. *Journal of Applied Econometrics*, 32(6):1197–1205, 2017.

Thomas J. Biersteker, Sue E. Eckert, and Marcos Tourinho, editors. *Targeted Sanctions: The Impacts and Effectiveness of United Nations Action*. Cambridge University Press, Cambridge, 2016.

Jiafeng Chen and Jonathan Roth. Logs with zeros? Some problems and solutions. *Quarterly Journal of Economics*, 139(2):891–936, 2024.

Michael Coppedge, John Gerring, Carl Henrik Knutsen, et al. V-Dem country-year dataset. Varieties of Democracy (V-Dem) Project, version 15, 2025.

Shawn Davies, Therese Pettersson, and Magnus Öberg. Organized violence 1989–2025, and violent political protests. *Journal of Peace Research*, 63(4):705–723, 2026. doi: 10.1093/jopres/xjag046.

Clément de Chaisemartin and Xavier D'Haultfœuille. Difference-in-differences estimators of intertemporal treatment effects. arXiv:2007.04267, version 14; also NBER Working Paper 29873, 2026.

Geert Dhaene and Koen Jochmans. Split-panel jackknife estimation of fixed-effect models. *Review of Economic Studies*, 82(3):991–1030, 2015.

Sajjad F. Dizaji and Mohammad Reza Farzanegan. Democracy and militarization in developing countries: A panel vector autoregressive analysis. *Defence and Peace Economics*, 34(3):272–292, 2023.

Sajjad F. Dizaji and Syed Mansoob Murshed. External arms embargoes and their implications for government expenditure, democracy and internal conflict. *World Development*, 173:106410, 2024.

Daniel W. Drezner. Sanctions sometimes smart: Targeted sanctions in theory and practice. *International Studies Review*, 13(1):96–108, 2011.

John C. Driscoll and Aart C. Kraay. Consistent covariance matrix estimation with spatially dependent panel data. *Review of Economics and Statistics*, 80(4):549–560, 1998.

Arindrajit Dube, Daniele Girardi, Òscar Jordà, and Alan M. Taylor. A local projections approach to difference-in-differences. NBER Working Paper 31184, National Bureau of Economic Research, 2023.

Elena-Ivona Dumitrescu and Christophe Hurlin. Testing for Granger non-causality in heterogeneous panels. *Economic Modelling*, 29(4):1450–1460, 2012.

Abel Escribà-Folch. Economic sanctions and the duration of civil conflicts. *Journal of Peace Research*, 47(2):129–141, 2010.

Gabriel Felbermayr, Aleksandra Kirilakha, Constantinos Syropoulos, Erdal Yalcin, and Yoto V. Yotov. The global sanctions data base. *European Economic Review*, 129:103561, 2020.

Kristian S. Gleditsch and Michael D. Ward. Interstate system membership: A revised list of the independent states since 1816. *International Interactions*, 25(4):393–413, 1999.

Lars Peter Hansen. Large sample properties of generalized method of moments estimators. *Econometrica*, 50(4):1029–1054, 1982.

Douglas Holtz-Eakin, Whitney Newey, and Harvey S. Rosen. Estimating vector autoregressions with panel data. *Econometrica*, 56(6):1371–1395, 1988.

Gary Clyde Hufbauer, Jeffrey J. Schott, Kimberly Ann Elliott, and Barbara Oegg. *Economic Sanctions Reconsidered*. Peterson Institute for International Economics, Washington, DC, 3rd edition, 2007.

Lisa Hultman and Dursun Peksen. Successful or counterproductive coercion? The effect of international sanctions on conflict intensity. *Journal of Conflict Resolution*, 61(6):1315–1339, 2017. doi: 10.1177/0022002715603453.

David A. Jaeger and M. Daniele Paserman. The cycle of violence? An empirical analysis of fatalities in the Palestinian-Israeli conflict. *American Economic Review*, 98(4):1591–1604, 2008.

Òscar Jordà. Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182, 2005.

Artūras Juodis, Yiannis Karavias, and Vasilis Sarafidis. A homogeneous approach to testing for Granger non-causality in heterogeneous panels. *Empirical Economics*, 60:93–112, 2021.

Kalev Leetaru and Philip A. Schrodt. GDELT: Global data on events, location, and tone, 1979–2012. In *Annual Convention of the International Studies Association*, San Francisco, 2013.

David Lektzian and Patrick M. Regan. Economic sanctions, military interventions, and civil conflict outcomes. *Journal of Peace Research*, 53(4):554–568, 2016.

Dake Li, Mikkel Plagborg-Møller, and Christian K. Wolf. Local projections vs. VARs: Lessons from thousands of DGPs. *Journal of Econometrics*, 244(2):105722, 2024.

Inessa Love and Lea Zicchino. Financial development and dynamic investment behavior: Evidence from panel VAR. *Quarterly Review of Economics and Finance*, 46(2):190–210, 2006.

Ziwei Mei, Liugang Sheng, and Zhentao Shi. Nickell bias in panel local projection: Financial crises are worse than you think. arXiv:2302.13455, version 5, 2026.

Steven V. Miller. peacesciencer: An R package for quantitative peace science research. *Conflict Management and Peace Science*, 39(6):755–779, 2022.

Saban Nazlioglu and Cagin Karul. Testing for Granger causality in heterogeneous panels with cross-sectional dependence. *Empirical Economics*, 67:1541–1579, 2024.

Stephen Nickell. Biases in dynamic models with fixed effects. *Econometrica*, 49(6):1417–1426, 1981.

Dursun Peksen. Better or worse? The effect of economic sanctions on human rights. *Journal of Peace Research*, 46(1):59–77, 2009.

Dursun Peksen and A. Cooper Drury. Coercive or corrosive: The negative impact of economic sanctions on democracy. *International Interactions*, 36(3):240–264, 2010.

M. Hashem Pesaran. A simple panel unit root test in the presence of cross-section dependence. *Journal of Applied Econometrics*, 22(2):265–312, 2007.

M. Hashem Pesaran. Testing weak cross-sectional dependence in large panels. *Econometric Reviews*, 34(6–10):1089–1117, 2015.

Mikkel Plagborg-Møller and Christian K. Wolf. Local projections and VARs estimate the same impulse responses. *Econometrica*, 89(2):955–980, 2021.

David Roodman. A note on the theme of too many instruments. *Oxford Bulletin of Economics and Statistics*, 71(1):135–158, 2009.

J. M. C. Santos Silva and Silvana Tenreyro. The log of gravity. *Review of Economics and Statistics*, 88(4):641–658, 2006.

E. Stork. Pressuring rebels: A quantitative study on sanctions' effects on violence against civilians. Bachelor's thesis, Peace and Conflict Studies, Uppsala University, 2020.

Joaquin Urtuzuastigui and Ore Koren. Targeted sanctions, resource substitution, and violence against civilians: Localized evidence from African states. *World Development*, 179:106606, 2024.

Reed M. Wood. "A hand upon the throat of the nation": Economic sanctions and state repression, 1976–2001. *International Studies Quarterly*, 52(3):489–513, 2008.

Jeffrey M. Wooldridge. *Econometric Analysis of Cross Section and Panel Data*. MIT Press, Cambridge, MA, 2nd edition, 2010.

Jiaqi Xiao, Artūras Juodis, Yiannis Karavias, Vasilis Sarafidis, and Jan Ditzen. Improved tests for Granger noncausality in panel data. *Stata Journal*, 23(1):230–242, 2023.

Erdal Yalcin, Gabriel Felbermayr, Hamid Kariem, Aleksandra Kirilakha, Ohyun Kwon, Constantinos Syropoulos, and Yoto V. Yotov. The global sanctions data base – release 4: The heterogeneous effects of the sanctions on Russia. WIFO Working Paper 681, WIFO, 2024.

A Additional tables and figures

Table 15: Reaction functions (FE-OLS): all coefficients

	Sender equation		Violence equation	
Dependent variable	New sanction	New cases	Conflict	log deaths
A. With lagged controls				
Violence, $t-1$	0.029^{*}	0.028	0.352^{***}	0.585^{***}
	(0.017)	(0.024)	(0.030)	(0.046)
Violence, $t-2$	0.027	0.002	0.133^{***}	0.062
	(0.018)	(0.023)	(0.028)	(0.039)
New sanctions, $t-1$	0.130^{***}	0.086^{***}	-0.000	-0.007
	(0.018)	(0.020)	(0.013)	(0.015)
New sanctions, $t-2$	-0.016	-0.007	-0.035^{***}	-0.044^{***}
	(0.017)	(0.015)	(0.013)	(0.016)
Sanction stock, $t-1$	-0.002	-0.023^{*}	0.009^{**}	0.002
	(0.005)	(0.014)	(0.004)	(0.008)
GDP growth, $t-1$	-0.365^{**}	-0.984^{***}	-0.380^{***}	-1.028^{***}
	(0.154)	(0.350)	(0.106)	(0.279)
Democracy, $t-1$	-0.155^{**}	-0.433^{***}	-0.111	-0.054
	(0.060)	(0.150)	(0.077)	(0.111)
B. State and year FE only				
Violence, $t-1$	0.037^{**}	0.047^{**}	0.365^{***}	0.605^{***}
	(0.017)	(0.022)	(0.030)	(0.044)
Violence, $t-2$	0.032^{*}	-0.002	0.142^{***}	0.057
	(0.018)	(0.024)	(0.028)	(0.038)
New sanctions, $t-1$	0.130^{***}	0.074^{***}	0.020^{*}	0.002
	(0.017)	(0.021)	(0.012)	(0.015)
New sanctions, $t-2$	-0.017	-0.018	-0.018	-0.037^{**}
	(0.017)	(0.018)	(0.012)	(0.015)

Notes: Bivariate panel VAR(2), equation by equation, state and year fixed effects, standard errors clustered by state in parentheses. Columns (1)–(2): new sanctions on lags of violence and of new sanctions; (3)–(4): violence on lags of violence and of new sanctions. 1991–2019. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 16: Granger tests for different lag structures: panel version of Jaeger and Paserman (2008), Table 2

	Violence → sanctions		Sanctions → violence	
Lags (own, other)	Incidence	Levels	Incidence	Levels
A. With lagged controls				
(1, 1)	$0.010^{**\dagger}$	$0.046^{**\dagger}$	0.862	0.361
(2, 1)	0.010^{**}	0.047^{**}	0.977	0.608
(3, 1)	0.011^{**}	0.048^{**}	$0.985^{\dagger}$	0.759
(1, 2)	0.028^{**}	0.137	0.104	0.028^{**}
(2, 2)	0.028^{**}	0.139	0.029^{**}	0.018^{**}
(3, 2)	0.029^{**}	0.140	0.031^{**}	$0.026^{**\dagger}$
(1, 3)	0.064^{*}	0.240	0.210	0.051^{*}
(2, 3)	0.063^{*}	0.242	0.066^{*}	0.029^{**}
(3, 3)	0.067^{*}	0.248	0.064^{*}	0.041^{**}
B. State and year FE only				
(1, 1)	$0.002^{***\dagger}$	$0.005^{***\dagger}$	0.075^{*}	0.631
(2, 1)	0.002^{***}	0.005^{***}	0.124	0.793
(3, 1)	0.002^{***}	0.004^{***}	$0.139^{\dagger}$	0.853
(1, 2)	0.006^{***}	0.015^{**}	0.176	0.081^{*}
(2, 2)	0.006^{***}	0.014^{**}	0.102	0.031^{**}
(3, 2)	0.005^{***}	0.014^{**}	0.101	$0.040^{**\dagger}$
(1, 3)	0.016^{**}	0.027^{**}	0.192	0.167
(2, 3)	0.015^{**}	0.025^{**}	0.149	0.069^{*}
(3, 3)	0.013^{**}	0.027^{**}	0.185	0.086^{*}

Notes: p-values of the Granger test (FE-OLS, clustered by state) for every combination of own lags and lags of the other variable; common sample with three lags of both variables (1992–2019), so that BIC is comparable. † lowest BIC in the column. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

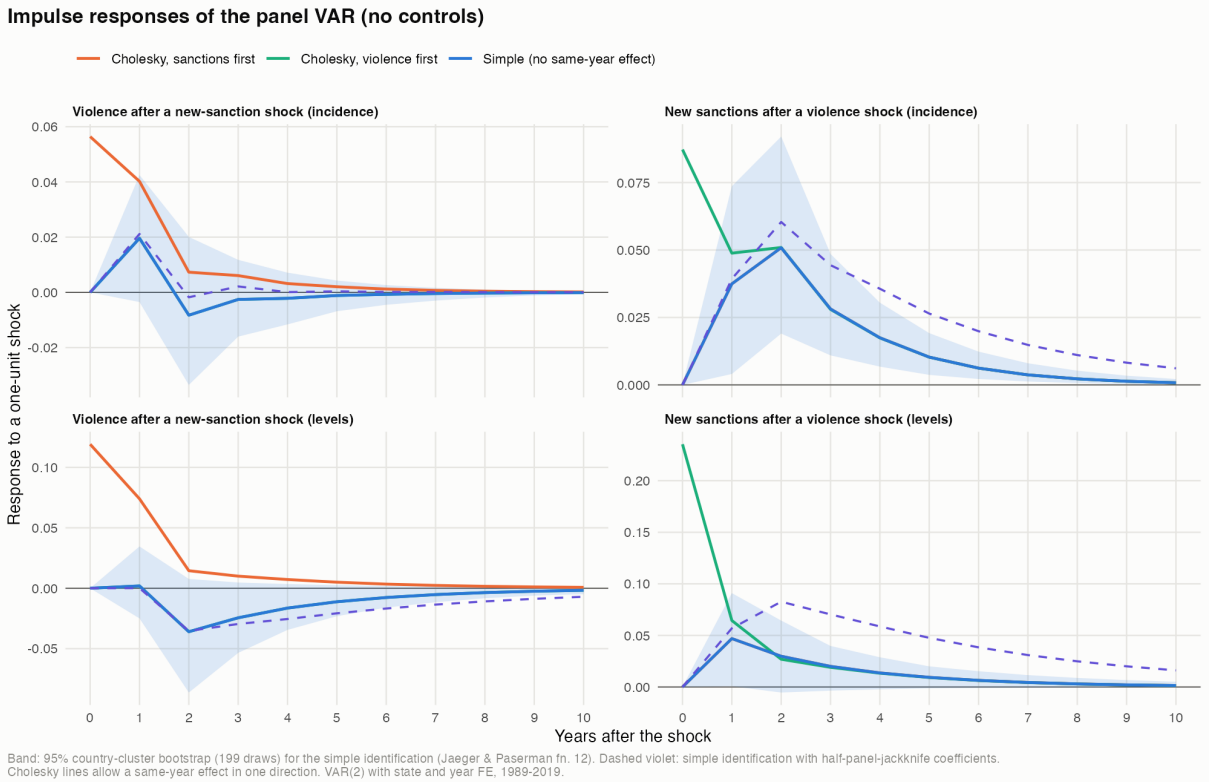


Figure 10: Impulse responses, VAR without controls

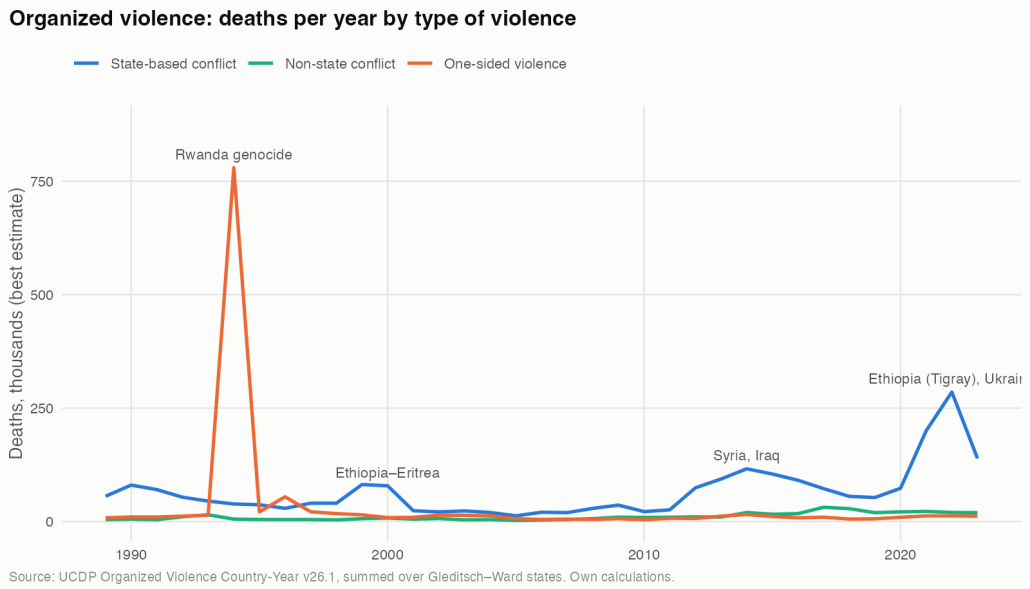


Figure 11: Deaths in organized violence by type (UCDP)